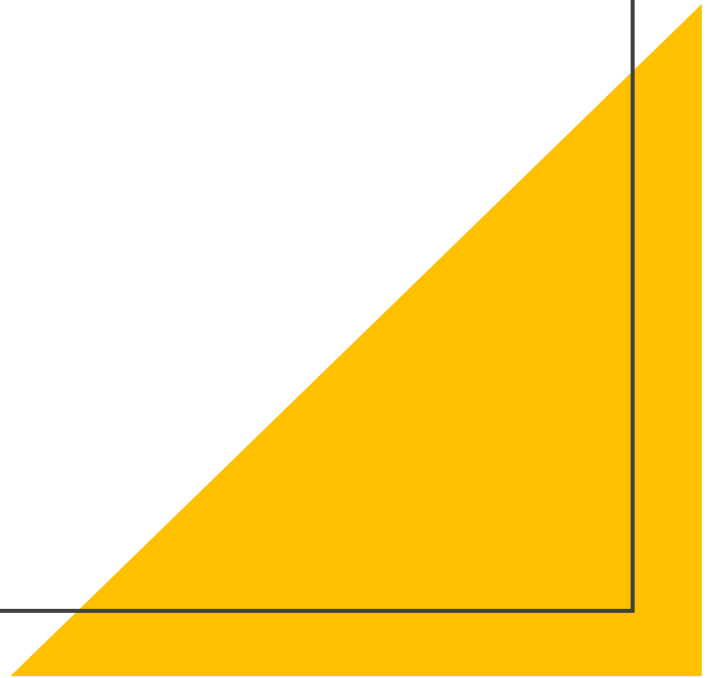


Sociotechnical Security

CS 8803

Professor Veronica Rivera

Lecture 1 – Welcome!



Technical system failures



2014: Hundred of celebrities' nude photos stolen and leaked



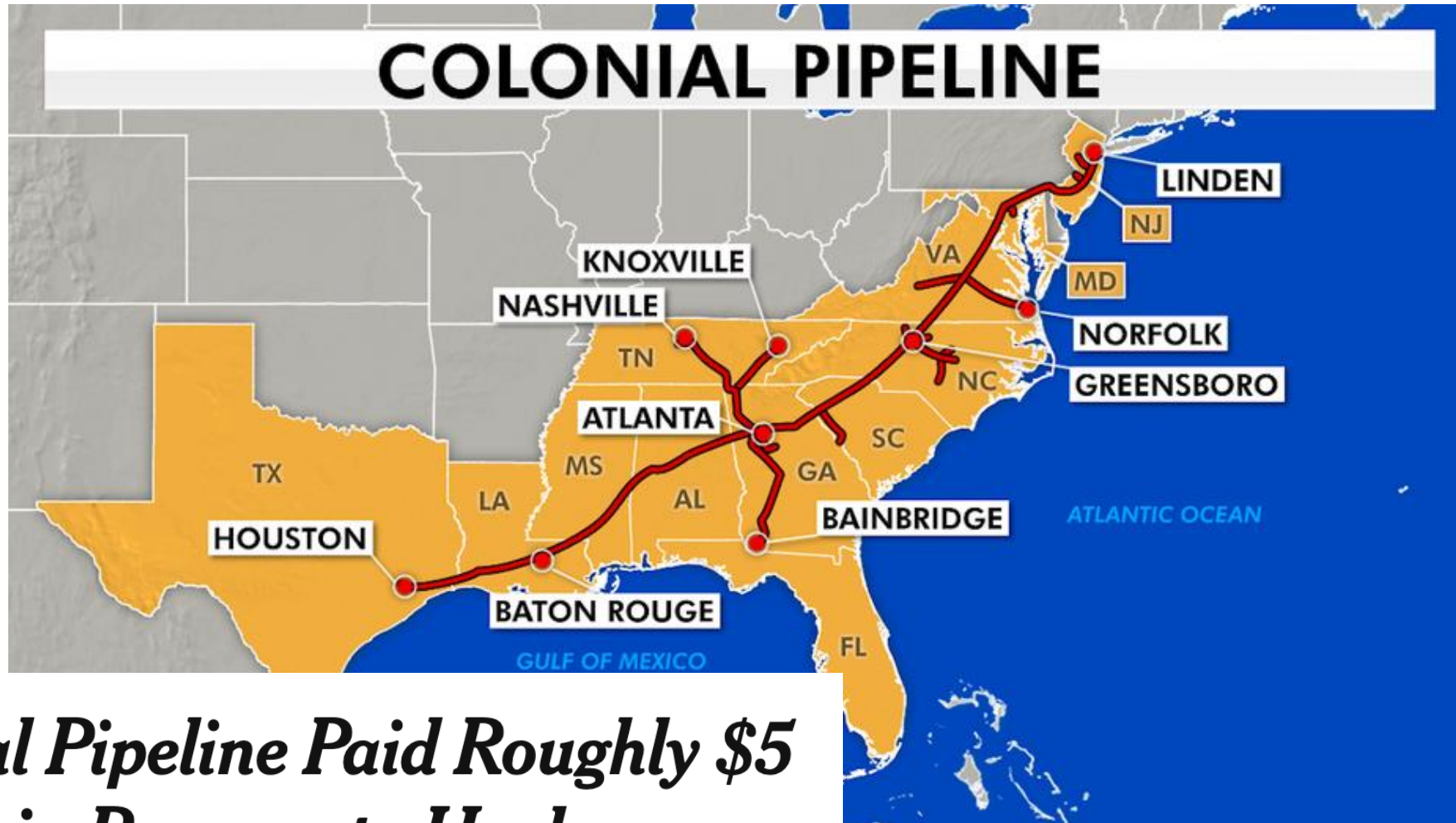
2017: Personal information of millions of US consumers stolen in one of the largest data breaches in history



Life360

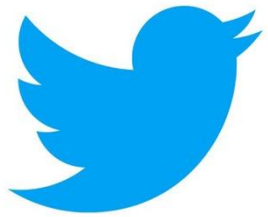
2024: Names, phone numbers, and emails of nearly 500,000 users were exposed

People using secure systems insecurely



Colonial Pipeline Paid Roughly \$5 Million in Ransom to Hackers

Exploiting human in the loop



Barack Obama ✓ @BarackObama

I am giving back to m

All Bitcoin sent to my
\$1,000, I will send ba



Apple ✓
@Apple

We are giving back to our community.
We support Bitcoin and we believe you
should too!

All Bitcoin sent to our address below
will be sent back to you doubled!

[bc1qxy2kgdvgmstus2n0yrt2403p83u](#)
[k0j0u0vahn](#)

Only going on for the next 30 minutes.

21:58 · 15/07/2020 · Twitter Web App

The Trump Administration Accidentally Texted Me Its War Plans

U.S. national-security leaders included me in a group chat about
upcoming military strikes in Yemen. I didn't think it could be real.
Then the bombs started falling.

All Bitcoin sent to my address below will
be sent back doubled. I am only doing a
maximum of \$50,000,000.

[bc1qxy2kgdvgmstus2n0yrt2403p83u](#)
[k0j0u0vahn](#)

Enjoy!

788

1,480

1,771



Trusted systems doing untrustworthy things

50 million Facebook profiles harvested for Cambridge Analytica in major data breach



Fitness tracking app Strava gives away location of secret US army bases

Data about exercise routes shared online by soldiers can be used to pinpoint overseas facilities

- Latest: Strava suggests military users 'opt out' of heatmap as row deepens

Dual-use systems & weaponization



Two women who allege they were stalked and harassed using AirTags are suing Apple



Dover woman loses \$15K after scammers used artificial intelligence to impersonate daughter

By Aaron Mesmer | FOX 13 News | Hillsborough County

"When I picked up the phone, it was my daughter's voice," Brightwell recalled. "It was her crying voice, she was hysterical."

How Gamergate foreshadowed the toxic hellscape that the internet has now become

Poor design is the root of many system failures



Technical



Mistake



Deception

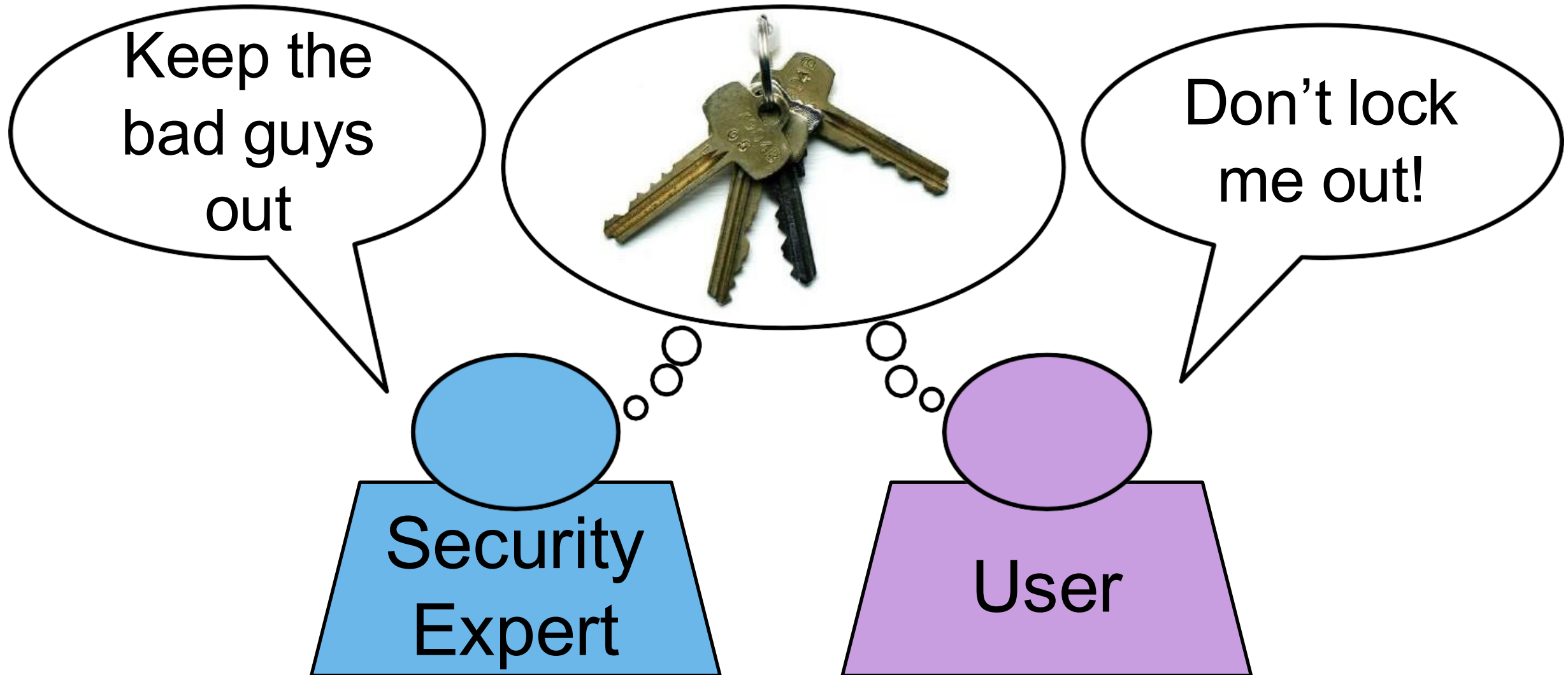


Trust



Misuse

Concerns may not be aligned



Computer security doesn't stop at
protecting technical systems; we must also
protect **people**

Protecting people = sociotechnical security

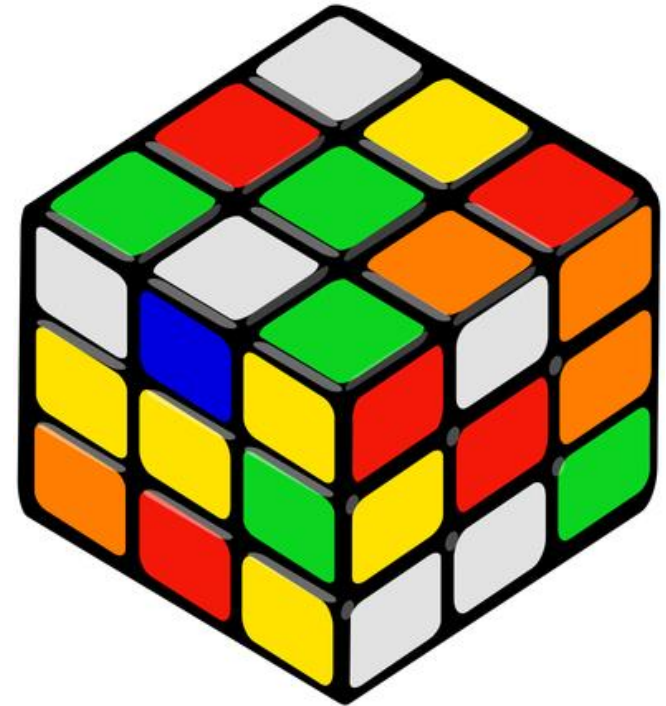
**keep people safe
in their interactions with sociotechnical
systems**

Usable security and
privacy

Digital safety

Problems are hard to solve (but interesting!)

- Harms are subjective
 - No universal fix
- Harms are rooted in societal challenges
 - Must consider these other dimensions
- Harms cannot be mathematically formalized
 - Need to use other techniques to solve

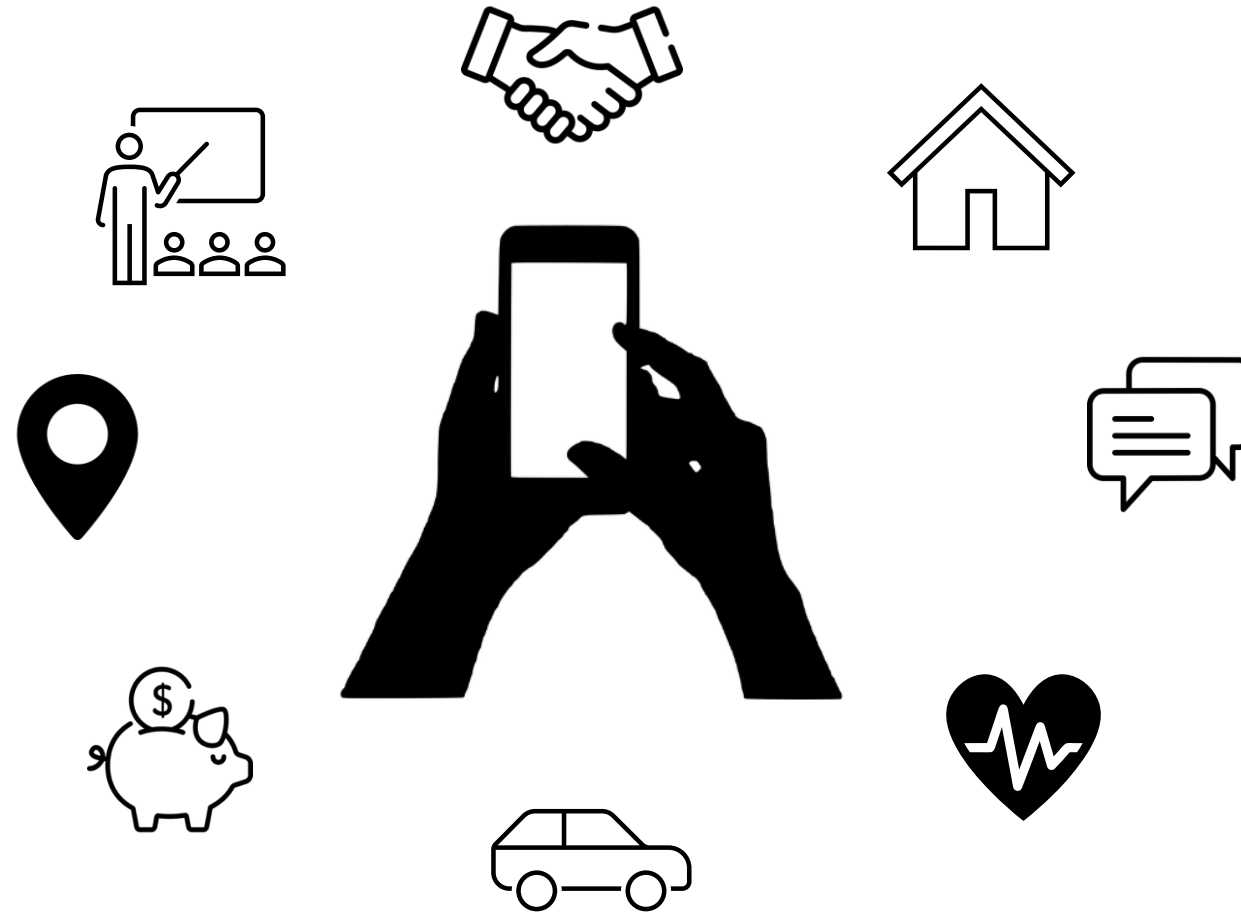


Studying human behavior is key

We can borrow models and methods from other disciplines that study human behavior:

- Human-computer interaction
- Psychology
- Sociology
- Ethnography
- Cognitive sciences
- Behavioral economics
- Warnings science
- Risk perception
- Organizational change
- Marketing
- Counterterrorism
- Communication
- Persuasive technology
- Learning science
- Network analytics

Lots of opportunity for real-world impact



You may also hear the term “trust and safety”

“Trust and Safety is the study of how the technology platforms that facilitate interactions between human beings can be used to cause harm and the steps that can be taken to prevent or mitigate those harms.”

- Alex Stamos, Shelby Grossman, and Riana Pfefferkorn
(Trust and Safety: An Open Source Textbook)

Lots of roles in trust and safety

Applied Research Scientist - Trust and Safety

TikTok [🔗](#) | New York, NY | \$219,000 - \$408,000 a year

You must create an Indeed account before continuing to the company website to

[Apply on company site](#) [🔗](#)



Responsibilities

The Applied Research Scientist will drive rigorous research to develop a deep understanding of complex Trust & Safety challenges and inform better policy, product, and operational decisions. This role combines quantitative analysis, qualitative research, and domain expertise to uncover emerging risks, evaluate safety interventions, and translate insights into actionable strategies. The researcher will partner closely with Data Science, Policy, Product, Operations, and Engineering teams to shape how we understand and address evolving safety challenges at scale.

Responsibilities: Conduct Mixed-Methods Research on Complex Safety Problems:

- Lead end-to-end research initiatives, from problem framing and hypothesis development to analysis, validation, and recommendations.
- Apply quantitative methods including statistical analysis, experimentation, causal inference, and modeling to identify trends, measure impact, and evaluate interventions.
- Apply qualitative methods including user research, case analysis, stakeholder interviews, and ecosystem research to understand behaviors, motivations, and emerging risks.

Engineering Analyst, Trust and Safety, Egregious Harms

Google [🔗](#) | Washington, DC | \$132,000 - \$189,000 a year

You must create an Indeed account before continuing to the company website to

[Apply on company site](#) [🔗](#)



Trust and Safety team members are tasked with identifying and solving problems that challenge the safety and integrity of our products. They use their know-how, excellent problem-solving skills, user insights, and communication to protect users and our partners from abuse across Google products like Search, Maps, Gmail, and Google Ads. On this team, you're a big-picture thinker and strategic team-player with a passion for doing what's right. You work globally and cross-functionally with Google engineers and product managers to identify and fight abuse and fraud cases at Google speed - with urgency. And you take pride in knowing that every day you are working hard to promote trust in Google and ensuring the highest levels of user safety.

As an Engineering Analyst within the Trust and Safety team, you will be part of a global team that ensures the seamless detection and enforcement of some of the most egregious online harms (e.g., child sexual abuse materials, non-consensual intimate imagery, violent extremism). Through close partnerships with Product, Engineering, Policy, and Legal, you will leverage your technical acumen to design and develop novel solutions to further Google's ability to fight this abuse worldwide. At Google we work hard to earn our users' trust every day. Trust and Safety is Google's team of abuse fighting and user trust experts working daily to make the internet a safer place. We partner with teams across Google to deliver bold solutions in abuse areas such as

I'm in, tell me more!



Welcome to sociotechnical security

Course objectives:

1. Speak knowledgably about topics at the intersection of human behavior, security & privacy, and technology design
2. Identify potential failures of digital systems through a human-centered lens
3. Evaluate the impact of digital systems on individual and societal-level security, privacy, and safety
4. Apply human-centered and empirical research methods to design for and evaluate security, privacy, and safety in digital systems

Professor Rivera (she/her)

Academic researcher in human-centered security & privacy



BS in CS/Math
'17



PhD in Computational
Media '23



Postdoctoral researcher in
CS + Ethics ('23-'25)



Assistant Professor



Postdoctoral
researcher

What brings **you** here?

- Tell us your name, degree program, and how long you've been in Atlanta
- Why you're taking this class
- Something you're looking forward to this semester (doesn't have to be class related!)

Course outline

Weeks 1-2 (Introduction and what defines the field)

- Introduction to sociotechnical security
- High-level introduction to research methods in sociotechnical security
- User mental models
- Abuse vectors

Weeks 3-6 (Usable security & privacy)

- Authentication
- Security warnings
- Privacy
- Phishing and social engineering

Course outline

Weeks 7-14 (Digital safety)

- Scams
- Online hate and harassment
- Intimate partner violence
- S&P of activists and journalists
- Digital intimacy
- Youth safety
- AI safety
- Demographic and cultural considerations in S&P

Weeks 15-16 (Ethics & careers in sociotechnical security)

- Ethical considerations
- Careers
- Final project presentations

Weekly/daily structure

We will spend 1 week on most topics.

Day 1

Lecture:

- 45-50 minutes

Discussion or class activity:

- 25-30 minutes

Day 2

Discussion:

- 75 minutes

Course resources

Course schedule, assignments, and all materials will be posted on the course website: <https://sociotechnical-security.github.io/>

BOOKMARK THIS PAGE!!

Assignments are submitted on Gradescope:

- Link: <https://www.gradescope.com/courses/1365210>
- Join code: 68V7YV

Announcements will be posted on Ed:

- Join link: <https://edstem.org/us/join/jvrPr5>
- Direct link: <https://edstem.org/us/courses/102980/discussion>

Assignments

There are 3 types of assignments in the course:

Reading responses

- Due at the start of each class period, beginning week 2
- Readings will be posted on website at least two weeks before they are due

Leading discussion

- Each student does this twice, beginning week 4
- No reading response due when you are leading discussion
- I will assign you to two presentation slots by next week.

Semester-long project

- Completed in groups of 1-3, project proposal due in week 4. **But start thinking about your project now, don't wait!**

Project

- The most significant assignment in this course is the semester project, which will be completed in groups of 1-3 of your choosing.
 - Intention is to give you experience getting a flavor for research in this area
- Two options:
 - **Open-ended research project** on a sociotechnical security topic using any human-centered and/or empirical research methods
 - **Platform harms measurement project**, building an automated detector for identifying harm on a social platform and using the tool to answer a set of research questions

Project Milestones

- Project proposal: Due 9/14
 - Detailed instructions for project proposal will be on the course website by next Monday.
- Midpoint check-in report: Due 10/14
- Final presentation: 12/7 (last day of class)
- Final report: Due 12/7

There are ample opportunities for you to receive help from Veronica on the project.

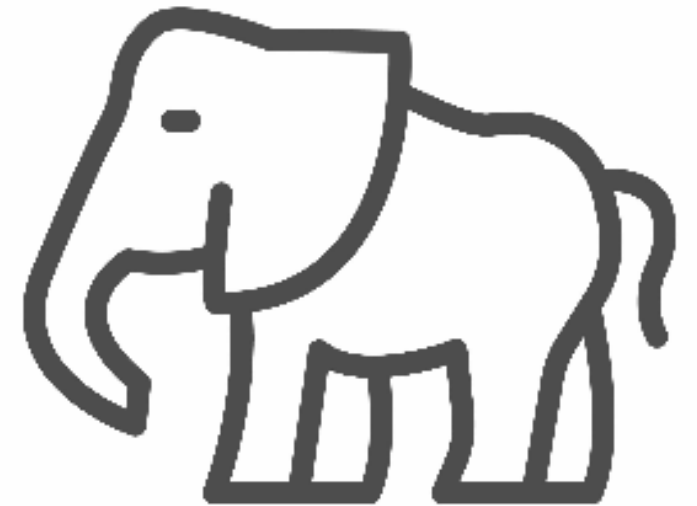
Grading

- Attendance and participation: 15%
- Reading response: 10%
 - No late reading responses accepted. I will drop your lowest 2 grades.
- Leading discussion: 25%
- Semester project: 50%

Addressing the elephant in the room (AI)

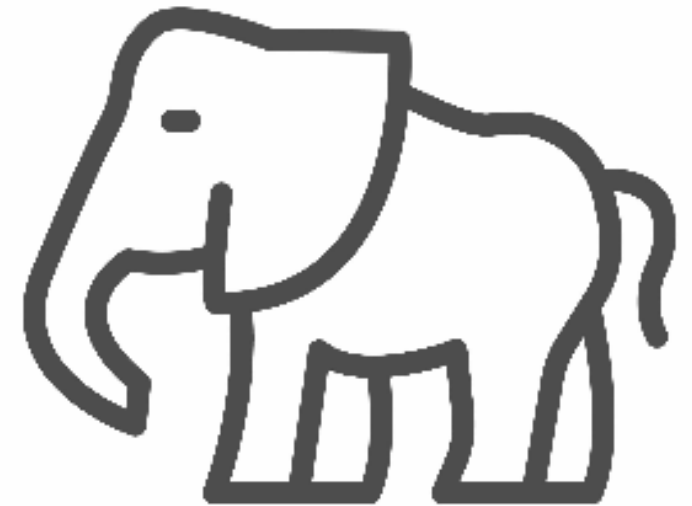
For reading responses and leading discussion, you can use generative AI tools **only** to check spelling and grammar and as you would a search engine. For the latter, **always** double check AI's responses.

You **cannot** use AI to do the thinking for you (e.g., to generate any deliverables for the course or to summarize papers for you). **If you use generative AI to summarize papers instead of reading them yourself, I will be able to tell during the discussion and this will be reflected in your participation grade.**



One caveat:

For the course project, you **can** use LLMs as a coding **assistant**, but you if you are building a system as your final project, **you are ultimately responsible for understanding the design decisions for your system and for explaining how your system works. You will be asked questions about this during the final project presentation.**



Administrative

Office hours will be Wednesday 4-5 PM, or by appointment, in the Coda second floor common space.

If you don't have access to the elevators using your buzzcard, use the escalators next to Costa Coffee.

You don't need to have a specific question about the course to come talk to me! I'm always happy to think through your ideas or talk about research/careers in human-centered security and privacy.

Contact

Email (best option): varivera@gatech.edu. Add [CS 8803] to the subject.

Office: Coda 1023

- If you don't already have Coda access, but want to meet me in my office, let me know ahead of time so I can make plans to get you up the elevators.

Disability accommodations

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services (<https://disabilityservices.gatech.edu>; 404-894-2563) as soon as possible to make an appointment to discuss your special needs and to obtain an accommodations letter.

Please also e-mail me as soon as possible in order to set up a time to discuss your learning needs.

A note on content



We will cover real-world harmful behavior and potentially upsetting materials

These topics include, but are not limited to: interpersonal abuse, hate speech, harassment, stalking, child and adult sexual exploitation, and more.

We need to understand these harms before we can stop them.

Please take care of yourself. You can step out of class at any point if you feel uncomfortable or need a break.

Activity time!

Instructions

Recall the 5 categories at the root of many computer system failures:

1. **Technical:** Bug in code
2. **Mistake:** System is not designed to reflect human behavior
3. **Deception:** A human in the loop is tricked
4. **Trust:** System is designed to tradeoff user well-being against another value
5. **Misuse:** System is weaponized to cause harm

- Identify a news headline
- Research the story behind it (what happened, what was the impact, what was the resolution)
- Identify which category or categories your examples fall into and why.
- Be prepared to share out with the rest of the class

Next time: research methods overview

- Short reading posted on course website
- No reading response
- If you have read research papers (in any field) before, come prepared to share what you have found challenging/helpful about this. If you have not read research papers before, come ready to learn from your peers!